Linear Programming Examples 1

Lecture 3

ETF2480&ETF5248 - S2, 2026
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Today's Goal



- The Diet Problem (and product mix)
- Dynamic (multi-period) Problems
- Production Process

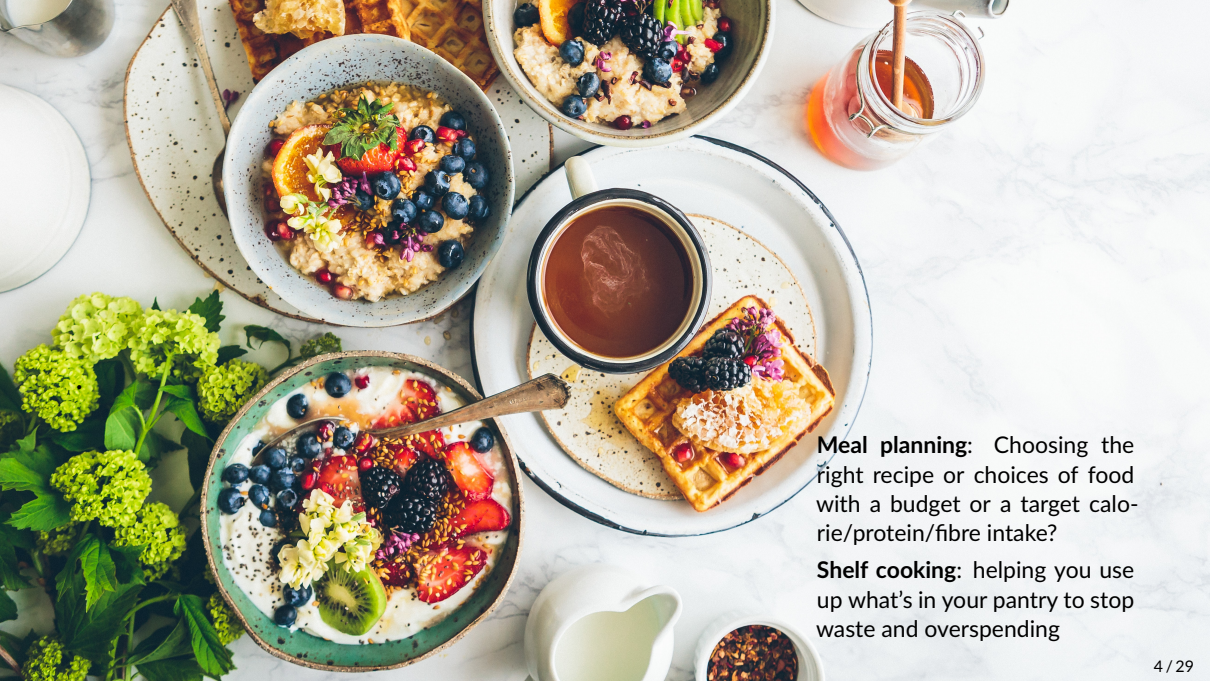
How are they related to Businesses?

These problems occur in different parts of a business in different scenarios, taking different forms or shapes. Recognising these are LPs and therefore can be solved efficiently is key.

Have you used a meal planning app?

What do you use it for?





Meal planning: Choosing the right recipe or choices of food with a budget or a target calorie/protein/fibre intake?

Shelf cooking: helping you use up what's in your pantry to stop waste and overspending

The Diet Problem

Diet problem: choose quantities $x_1, \ldots, x_n$ of n foods
one unit of food j costs c_j, contains amount a_{ij} of nutrient i
healthy diet requires nutrient i in quantity at least b_i

The goal is to find cheapest healthy diet.
Mathematically, the problem can be formulated as:

$$\begin{aligned} \text{minimize} \quad & c^T x \\ \text{subject to} \quad & Ax \geq b \end{aligned}$$

Several variations on this problem can also be formulated as LPs:

- can insist on an exact amount of a nutrient in the diet (which gives a linear equality constraint)
- can impose an upper bound on the amount of a nutrient
- etc.

The Diet Problem

Consider an instance of the diet optimisation problem:

- there are six different foods: bread, milk, cheese, potato, fish, and yogurt
- the cost and nutrition values per unit are displayed in the table below

The objective is to find a minimum-cost diet that contains

- at least 300 calories
- not more than 10 grams of protein
- not less than 10 grams of carbohydrates
- and not less than 8 grams of fat
- in addition, the diet should contain at least 0.5 unit of fish and no more than 1 unit of milk

	Bread	**Milk**	**Cheese**	**Potato**	**Fish**	**Yogurt**
Cost, $	2.0	3.5	8.0	1.5	11.0	1.0
Protein, g	4.0	8.0	7.0	1.3	8.0	9.2
Fat, g	1.0	5.0	9.0	0.1	7.0	1.0
Carbohydrates, g	15.0	11.7	0.4	22.6	0.0	17.0
Calories, cal	90	120	106	97	130	180

The Diet Problem: solution

Let x_j be the quantity of food j chosen. Then the diet problem can be formulated as

$$
\begin{aligned}
\text{minimize} \quad & 2x_1 + 3.5x_2 + 8x_3 + 1.5x_4 + 11x_5 + x_6 \\
\text{subject to} \quad & 90x_1 + 120x_2 + 106x_3 + 97x_4 + 130x_5 + 180x_6 \geq 300 \\
& 4x_1 + 8x_2 + 7x_3 + 1.3x_4 + 8x_5 + 9.2x_6 \leq 10 \\
& 15x_1 + 11.7x_2 + 0.4x_3 + 22.6x_4 + 17x_6 \geq 10 \\
& x_1 + 5x_2 + 9x_3 + 0.1x_4 + 7x_5 + x_6 \geq 8 \\
& x_5 \geq 0.5 \\
& x_2 \leq 1 \\
& x_j \geq 0, \quad j = 1, \ldots, 6
\end{aligned}
$$

solution: $x^\star = (0, 0.053, 0.449, 1.865, 0.500, 0)$ with $p^\star = 12.081$

Live Coding Time!

Let’s code the Diet problem together!

The Original Diet Problem

The diet problem was one of the first optimization problems studied in the 1930s and 1940s. The problem was motivated by the U.S. Army's desire to minimize the cost of feeding soldiers in the field while still providing a healthy diet. George Stigler, one of the early researchers to study the problem made an educated guess for the cost of an optimal diet was $39.93 per year (1939 prices).

Let's look at the data set. We will re-compute this optimal diet in the workshop. (Let's hope the meals from this diet is delicious.)

Product Mix

We've already seen a few product mix examples by now. The ApplePie (the company that sells laptops and desktops - lecture 1), the Giapetto's Woodcarving (selling wooden train and soldier - lecture 2), the Gurobi advertisements (marketing - lecture 2), and the diet problem are problems that solving for optimal mix of product to order/produce/prescribe.



Dynamic (multi-period) Problems

Often we need to make decisions for business operations for periods ahead.

Multi-period inventory

Sailco Corporation must determine how many sailboats should be produced during each of the next four quarters. Demand is known for the next four quarters. Sailco must meet demands on time. At the beginning of the first quarter Sailco has an inventory of 10 sailboats. Sailco must decide at the beginning of each quarter how many sailboats should be produced during each quarter.

During each quarter, Sailco can produce up to 40 sailboats with regular-time labor at a total cost of $400 per sailboat. By having employees work overtime during the quarter, Sailco can produce additional sailboats at a total cost of $450 per sailboat. At the end of each quarter a carrying or holding cost of $20 per sailboat is incurred.

Use LP to determine a production schedule to minimize the sum of production and inventory costs during the next four quarters.

Table: Demand for the next four quarters

Quarter	1	2	3	4
Demand	40	60	75	25

Multi-period inventory: solution

Define variables

- x_t - number of sailboats produced by regular-time labour during quarter t
- y_t - number of sailboats produced by overtime labour during quarter t
- i_t - number of sailboats on hand at the end of quarter t

Then the problem can be formulated as:

$$
\begin{aligned}
\text{minimize} \quad & 400(x_1 + \cdots + x_4) + 450(y_1 + \ldots y_4) + 20(i_1 + \cdots + i_4) \\
\text{subject to} \quad & i_1 = 10 + x_1 + y_1 - 40 \\
& i_2 = i_1 + x_2 + y_2 - 60 \\
& i_3 = i_2 + x_3 + y_3 - 75 \\
& i_4 = i_3 + x_4 + y_4 - 25 \\
& x_t \leq 40 \\
& x_t \geq 0, y_t \geq 0, i_t \geq 0, t = 1, \ldots, 4
\end{aligned}
$$

solution: $x^\star = (40, 40, 40, 25)$, $y^\star = (0, 10, 35, 0)$, $i^\star = (10, 0, 0, 0)$ with $p^\star = 78450$.

Multi-period investment (Finance)

Finco investment corporation must determine investment strategy for the firm during the next three years. Currently (time 0), $100,000 is available for investment. Investments A, B, C, D and E are available. The cash flow associated with investing $1 in each investment is given in the table below.

	cash flow ($) at time			
	0	1	2	3
A	-1	0.5	1	0
B	0	-1	0.5	1
C	-1	1.2	0	0
D	-1	0	0	1.9
E	0	0	-1	1.5

To ensure that the company's portfolio is diversified, Finco requires that at most $75,000 be placed in any single investment. In addition to investments A-E, Finco can earn interest at 8% per year by keeping uninvested cash in money market funds. Returns from investments may be immediately reinvested. Finco cannot borrow funds, so the cash available for investment at any time is limited to cash on hand. Formulate an LP that will maximise cash on hand at time 3.

Multi-period investment (Finance): solution

Let

x_i be the amount invested in investment i

s_t be the amount invested in money market funds at time t

Then the problem is:

$$
\begin{array}{rl}
\text{maximize} & x_B + 1.9x_D + 1.5x_E + 1.08s_2 \\
\text{subject to} & x_A + x_C + x_D + s_0 = 100000 \\
& 0.5x_A + 1.2x_C + 1.08s_0 = x_B + s_1 \\
& x_A + 0.5x_B + 1.08s_1 = x_E + s_2 \\
& x_i \leq 75000, \quad i = A, \ldots, E \\
& x_i \geq 0, \quad i = A, \ldots, E \\
& s_t \geq 0, \quad t = 0, \ldots, 2
\end{array}
$$

solution: $x^\star = (60000, 30000, 0, 40000, 75000)$, $s^\star = (0, 0, 0)$ with $p^\star = 218500$.

Multi-period work-scheduling (Human Resources Management)

A post office requires different number of employees on different days of the week (see Table). Union rules state that each employee must work five consecutive days and then receive two days off. For example, an employee who works Monday to Friday must be off on Saturday and Sunday. Formulate an LP that the post office can use to minimise the number of employees who must be hired.

Table: Requirements for post office

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
# of employees	17	13	15	19	14	16	11

Multi-period work-scheduling (Human Resources Management): *incorrect* formulation

Let x_i be the number of employees working on day i, then

$$
\begin{aligned}
\text{minimize} \quad & \sum_{i=1}^{7} x_i \\
\text{subject to} \quad & x_1 \geq 17 \\
& x_2 \geq 13 \\
& x_3 \geq 15 \\
& x_4 \geq 19 \\
& x_5 \geq 14 \\
& x_6 \geq 16 \\
& x_7 \geq 11 \\
& x_i \geq 0 \quad i = 1, \ldots, 7
\end{aligned}
$$

Multi-period work-scheduling (Human Resources Management): *correct* formulation

Let x_i be the number of employees beginning work on day i, then

$$
\begin{aligned}
\text{minimize} \quad & \sum_{i=1}^{7} x_i \\
\text{subject to} \quad & x_1 + x_4 + x_5 + x_6 + x_7 \geq 17 \\
& x_1 + x_2 + x_5 + x_6 + x_7 \geq 13 \\
& x_1 + x_2 + x_3 + x_6 + x_7 \geq 15 \\
& x_1 + x_2 + x_3 + x_4 + x_7 \geq 19 \\
& x_1 + x_2 + x_3 + x_4 + x_5 \geq 14 \\
& x_2 + x_3 + x_4 + x_5 + x_6 \geq 16 \\
& x_3 + x_4 + x_5 + x_6 + x_7 \geq 11 \\
& x_i \geq 0 \quad i = 1, \ldots, 7
\end{aligned}
$$

How to get integer solution?

But it doesn't make sense to have 0.33 employee to work in one day...

You are right! So what do you think we should do?

solution: $x^\star = (\frac{4}{3}, \frac{10}{3}, 2, \frac{22}{3}, 0, \frac{10}{3}, 5)$ with $p^\star = \frac{67}{3}$
rounded solution: $\tilde{x} = (2, 4, 2, 8, 0, 4, 5)$ with $\tilde{p} = 25$
integer solution: $x^\star_{ip} = (4, 4, 2, 6, 0, 4, 3)$ with $p^\star_{ip} = 23$

Let's come back to this in a few weeks (when we talk about Integer programming)! For now, keep your LP optimal solutions x^* and p^* to 4 decimal places in your assessments.

Production Process

Production process of good and services in different lines of business can be modelled as an LP.

Blending or Mixing

Sunco Oil manufactures three types of gasoline (gas_1, gas_2, gas_3). Each type is produced by blending three types of crude oil ($crude_1$, $crude_2$, $crude_3$). The sales price per barrel of gasoline and the purchase price per barrel of crude oil are given in the table. Sunco can purchase up to 5000 barrels of each type of crude oil daily. Three types of gasoline differ in their octane rating and sulphur content. The crude oil blended to form gas_1 must have an average octane rating of at least 10 and contain at most 1% sulphur. The crude oil blended to form gas_2 must have an average octane rating of at least 8 and contain at most 2% sulphur. For gas_3 the values are 6 and 1% respectively. The octane rating and the sulphur content are given in the table. It costs $4 to transform one barrel of oil into one barrel of gasoline, and Sunco's refinery can produce up to 14000 barrels of gasoline daily. Sunco's customers require the following amounts of each gasoline daily: gas_1 - 3000 barrels; gas_2 - 2000 barrels; gas_3 - 1000 barrels. The company considers it an obligation to meet these demands. Sunco also has the option of advertising to stimulate demand for its products. Each dollar spent daily in advertising a particular type of gasoline increases the daily demand for that type by 10 barrels. Formulate an LP that will enable Sunco to maximise daily profits.

Blending or Mixing

Table: prices

gas	price, $	crude	price, $
1	70	1	45
2	60	2	35
3	50	3	25

Table: octane ratings and sulphur content

crude	octane rating	sulphur content, %
1	12	0.5
2	6	2.0
3	8	3.0

Blending or Mixing Solution

solution: let x_{ij} be the number of barrels of crude$_i$ used to produce gas$_j$ and let a_j be the amount of \$ spent on advertising gas$_j$

$$
\begin{aligned}
\text{maximize} \quad & 21x_{11} + 11x_{12} + x_{13} + 31x_{21} + 21x_{22} + 11x_{23} + 41x_{31} + 31x_{32} + 21x_{33} \\
& - a_1 - a_2 - a_3 \\
\text{subject to} \quad & x_{11} + x_{21} + x_{31} - 10a_1 = 3000 \\
& x_{12} + x_{22} + x_{32} - 10a_2 = 2000 \\
& x_{13} + x_{23} + x_{33} - 10a_3 = 1000 \\
& x_{11} + x_{12} + x_{13} \leq 5000 \\
& x_{21} + x_{22} + x_{23} \leq 5000 \\
& x_{31} + x_{32} + x_{33} \leq 5000 \\
& x_{11} + x_{12} + x_{13} + x_{21} + x_{22} + x_{23} + x_{31} + x_{32} + x_{33} \leq 14000 \\
& 12x_{11} + 6x_{21} + 8x_{31} \geq 10(x_{11} + x_{21} + x_{31}) \\
& 12x_{12} + 6x_{22} + 8x_{32} \geq 8(x_{12} + x_{22} + x_{32}) \\
& 12x_{13} + 6x_{23} + 8x_{33} \geq 6(x_{13} + x_{23} + x_{33}) \\
& 0.005x_{11} + 0.02x_{21} + 0.03x_{31} \leq 0.01(x_{11} + x_{21} + x_{31}) \\
& 0.005x_{12} + 0.02x_{22} + 0.03x_{32} \leq 0.02(x_{12} + x_{22} + x_{32}) \\
& 0.005x_{13} + 0.02x_{23} + 0.03x_{33} \leq 0.01(x_{13} + x_{23} + x_{33}) \\
& x_{ij} \geq 0, \quad a_i \geq 0, \quad i, j = 1, \ldots, 3
\end{aligned}
$$

Live Coding Time!

Let’s code the Blending problem together!

Perfumes Production Process

Rylon Corporation manufactures Brute and Chanelle perfumes. The raw material needed to manufacture each type of perfume can be purchased for $3 per pound. Processing 1 lb of raw material requires 1 hour of laboratory time. Each pound of processed raw material yields 3 oz of Regular Brute Perfume and 4 oz of Regular Chanelle Perfume. Regular Brute can be sold for $7/oz and Regular Chanelle for $6/oz.

Rylon also has the option of further processing Regular Brute and Regular Chanelle to produce Luxury Brute, sold at $18/oz and Luxury Chanelle, sold at $14/oz. Each ounce of Regular Brute precessed further requires an additional 3 hours of laboratory time and $4 processing cost and yields 1 oz of Luxury Brute. Each ounce of Regular Chanelle precessed further requires an additional 2 hours of laboratory time and $4 processing cost and yields 1 oz of Luxury Chanelle. Each year Rylon has 6000 hours of laboratory time available and can purchase up to 4000 lb of raw material.

Formulate an LP that can be used to determine how Rylon can maximise profits.

Assume that the cost of laboratory hours is a fixed cost.

Perfume Production Process Solution

- x_1 - number of ounces of Regular Brute sold annually
- x_2 - number of ounces of Luxury Brute sold annually
- x_3 - number of ounces of Regular Chanelle sold annually
- x_4 - number of ounces of Luxury Chanelle sold annually
- x_5 - number of pounds of raw material purchased annually

$$
\begin{aligned}
\text{maximize} \quad & 7x_1 + 14x_2 + 6x_3 + 10x_4 - 3x_5 \\
\text{subject to} \quad & x_5 \leq 4000 \\
& 3x_2 + 2x_4 + x_5 \leq 6000 \\
& x_1 + x_2 = 3x_5 \\
& x_3 + x_4 = 4x_5 \\
& x_i \geq 0, \quad i = 1, \ldots, 5
\end{aligned}
$$

Perfume Production Process Extension

Suppose that in the previous example, 1 lb of raw material could be used to produce either 3 oz of Brute **or** 4 oz of Chanelle. How would this change the formulation?

Add variables:

- x_6 - number of pounds of raw material to produce Brute
- x_7 - number of pounds of raw material to produce Chanelle

$$
\begin{aligned}
\text{maximize} \quad & 7x_1 + 14x_2 + 6x_3 + 10x_4 - 3x_5 \\
\text{subject to} \quad & x_5 \leq 4000 \\
& 3x_2 + 2x_4 + x_5 \leq 6000 \\
& x_1 + x_2 = 3x_6 \\
& x_3 + x_4 = 4x_7 \\
& x_6 + x_7 = x_5 \\
& x_i \geq 0, \quad i = 1, \ldots, 7
\end{aligned}
$$

solution: $x^\star = (0, 666.67, 15111.11, 0, 4000, 222.22, 3777.78)$ with $p^\star = 88000$

Quick Summary

We have covered 3 types of LPs today

1. Product Mix
2. Dynamic (multi-period) Problems
3. Production Process

Each of them is formulated differently and requires different modelling techniques.

These types of problem occur in different departments of a business in different scenarios, taking different forms or shapes. The important thing here is recognising these are actually LPs and therefore can be solved efficiently.

We will cover feasibility and the solving mechanism next week, where we will learn more about what is happening in R/Python under the hood.

Workshop: The original Stigler Diet Problem

Let's re-compute the solutions to the original Stigler Diet Problem in 1943.
Note that the units in the original problem is slightly different from the example we see during the lecture. In the Stigler problem, we divided each type of food into $1 portions, i.e. the amount of each type of food we can buy with $1. The nutrient columns are also rescaled to reflect this. For example, the amount of flour that I can get for $1 is 12600 gram and it contains 44.7 thousands of calories.
The csv files 'Stigler Food Info.csv' and 'Stigler Nutrient Requirement.csv' are provided to you alongside with the R code to execute them.
Discuss

- what operations does the R code perform?
- how did the mathematical formulation changed depending on the data that we have?
- what is the optimal solution and the corresponding diet?

Read more: Wikipedia page on Stigler Diet.

We will revisit the diet problem in various forms later in this unit.